

Assignment 2b – State Machine

01 Design

Scenario: Traffic Light

A simple finite state machine modeling the behavior of a traffic light. The light cycles automatically between three states using a timer, with each state having its own duration.

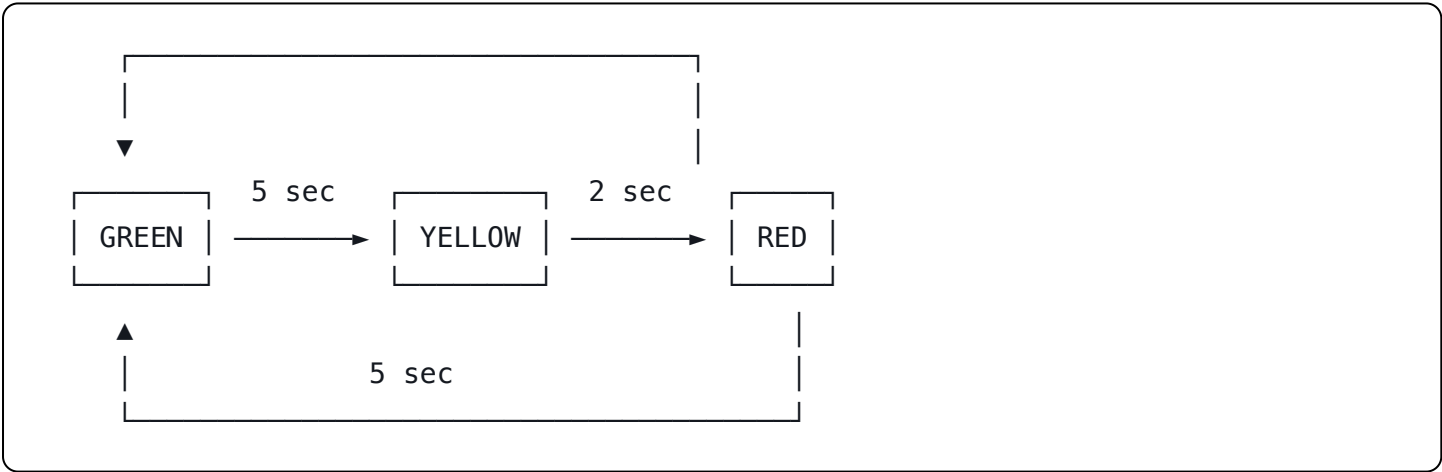
States

State	Color	Meaning
Green	Green	Vehicles are allowed to pass
Yellow	Yellow	Vehicles must slow down and prepare to stop
Red	Red	Vehicles must stop

State Transitions

Current State	Transition Condition	Next State
Green	After 5 seconds	Yellow
Yellow	After 2 seconds	Red
Red	After 5 seconds	Green

Transition Diagram



Notes

- The cycle is continuous: Green → Yellow → Red → Green → ...

- The yellow phase is intentionally shorter than the green and red phases, reflecting real-world traffic light behavior.
- All transitions are triggered automatically by a timer, with no external input required.
- Each state is visually distinguishable through a distinct color displayed on screen.